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Human Facial Emotion Detection Using Deep Learning



Dharma Karan Reddy Gaddam, Mohd Dilshad Ansari, Sandeep Vuppala,
Vinit Kumar Gunjan, and Madan Mohan Sati

Abstract Human emotions are important features in human communication. It is also equally important to recognize and classify emotions using computers, which drives this research to explore in various real world domains. Deep learning techniques are rapidly growing in this era and help in tackling real world problems. In this work, deep neural networks or convolutional neural networks are used to detect human facial emotions. We have proposed a network with the help of ResNet50 to classify the human facial emotions using static images. To train the network FER2013 dataset is used and the proposed model produced significant results compared to existing models.

Keywords Deep learning · Face detection · Human emotions · VGG · ResNet50 and AlexNet

1 Introduction

Facial emotion recognition is an active research area in the fields of computer vision and artificial intelligence. Numerous researchers are working in this research area over the past few decades, and it is still challenging due to complexities in this research domain [1]. Moreover, Emotion detection and recognition play a vital role in creating a huge impact in major real world applications. Verbal and non-verbal are two major forms of expressing an emotion. In verbal form, emotions are expressed in written form whereas in non-verbal form, emotions are expressed in facial emotion gestures and vocal tone of a person. A person can exhibit many expressions, where they can be classified into 7 basic emotions and 15 compound emotions. The basic emotions are happy, sad, surprise, anger, fear, disgust and neutral. In addition, the compound

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emotions are Happily surprised, Fearfully disgusted, Happily disgusted, Angrily surprised, Sadly fearful, Angrily disgusted, Sadly angry, Disgustedly surprised, Sadly surprised, Appalled, Sadly disgusted, Awed, Fearfully angry, Hatred, Fearfully surprised etc. CNN has been the most promising technique among various deep learning models. CNN is a class of deep neural networks, most commonly applied to analyze visual images and patterns from the images. Moreover, convolutional neural networks have produced noticeable results in most of the computer vision tasks and many other applications areas. CNN is able to produce higher accuracy over traditional machine learning models such as support vector machine (SVM), Neural Network etc. It can produce accurate results in different Complex conditions [1–10].

Many subsequent research works have been done in the field of computer vision. Human facial expressions are directly associated with facial muscles of humans. Features were extracted from the factors which are responsible for facial expressions and are trained on various machine learning or deep learning models to detect as well as classify human facial expressions. In this work we have proposed model using ResNet Architecture to classify the emotions, trained on the FER2013 dataset.

2 Related Work

Many approaches were developed for human facial recognition even before the outbreak of deep learning. These approaches include Local Binary Patterns (LBP), linear discriminant analysis (LDA), Principal component analysis (PCA), Locality preserving Projections (LPP) etc. Moreover, Local Binary Patterns (LBP) is used to find facial expressions automatically. This technique is employed to fix the position of eyes, eyeballs, and also focuses on face structure such as position of eyes, mouth and nearby areas on the face. After the evolution of machine learning, there is a lot of research going and different models have emerged such as support vector machines, neural networks approaches etc. [1]. Yang et al. [2] proposed Neural Network model, where vectorized facial features are used as input. The model can predict numerous emotions with the accuracy of 84.33%. Similarly, Liu [3] employed FER2013 dataset and a two and a two layer convolutional neural network model is used to identify different facial emotions. They compared their model with four different existing Methods and the proposed model achieved an accuracy of 49.8%. Additionally, Suresh et al. [4] developed a sign language recognition system which classifies six different sign languages using Deep Neural Networks. Adam and SGD optimizers are employed in two different approaches and compared them. Furthermore, the model using Adam optimizer performed good and produced significant results than SGD optimiser. Likewise, Verma and Tiwary [5] introduced an ensemble method using a support vector machine, Multilayer Perceptron, K-Nearest Neighbour and Meta-multiclass. The obtained average accuracy for SVM was 81.45%, MLP was 74.37%, KNN was 57.74% and MMC was 75.94%. Iacoviello et al. [6] presented a new hybrid classification model to detect the human emotions from EEG signals.

Fig. 1 No of sample images in each emotion of dataset



They combined SVM, principal component analysis (PCA) and discrete wavelet transformation [11–19].

3 Dataset

The dataset FER2013 is chosen for experimental analysis, which is an open-source dataset. It was first created for an on-going project by Pierre-Luc Carrier and Aaron Courville. Later this dataset was uploaded to kaggle for a competition. FER2013 dataset consists of 35,887 grayscale, 48×48 sized face images with various emotions. The emotions are labeled as (0: Angry, 1: Disgust, 2: Fear, 3: Happy, 4: Sad, 5: Surprise, 6: Neutral).

We can observe from the above Fig. 1 that there is strange variation in the number of samples of each emotion in the dataset. This variation will have an impact on the model and its efficiency. The dataset was in the form of a.csv excel sheet which comprises 48×48 pixel values of each image. Moreover, Images can be formed from the pixels of each image. Figure 2 is a sample of images of the FER2013 dataset, which is available at this link <https://www.kaggle.com/>.

In this dataset each image has less background grayscale images as color of person is not important to classify the emotion of a person and clear exposure of expression, this helps the model to extract more features with good efficiency.

4 Proposed Methodology

We proposed a deep learning model based on the convolutional neural network for classifying the human emotions based on static images. The complete network has different layers where images go through each layer of the network to classify its emotion. The layers are presented in Fig. 3.



Fig. 2 Greyscale images in fer2013

From the above diagram we can observe that input images are passed to the pre-processing stage. Moreover, Pre-processing is making data ready to pass it to the network such that the model predicts the output with high accuracy. Generally, in the Pre-processing stage noisy data is removed, uneven sizes of images are resized to a particular size, convert RGB to grayscale images.

And the next stage is Face Detection, as we are finding emotions of a human face it is required to find the face in an image. There could be multiple faces or no face in an image; therefore, it is significant to detect faces in an image. To detect faces MTCNN is used, MTCNN is an algorithm comprising 3 stages P, R, O, which detects bounding boxes of faces in an image. The bounding box is drawn based on coordinates in an image of a face. With the help of bounding boxes we can crop only faces in an image such that unnecessary parts in the image can be removed. The background is eliminated and left with only the face of an image. Now the data is good enough to pass to the model.

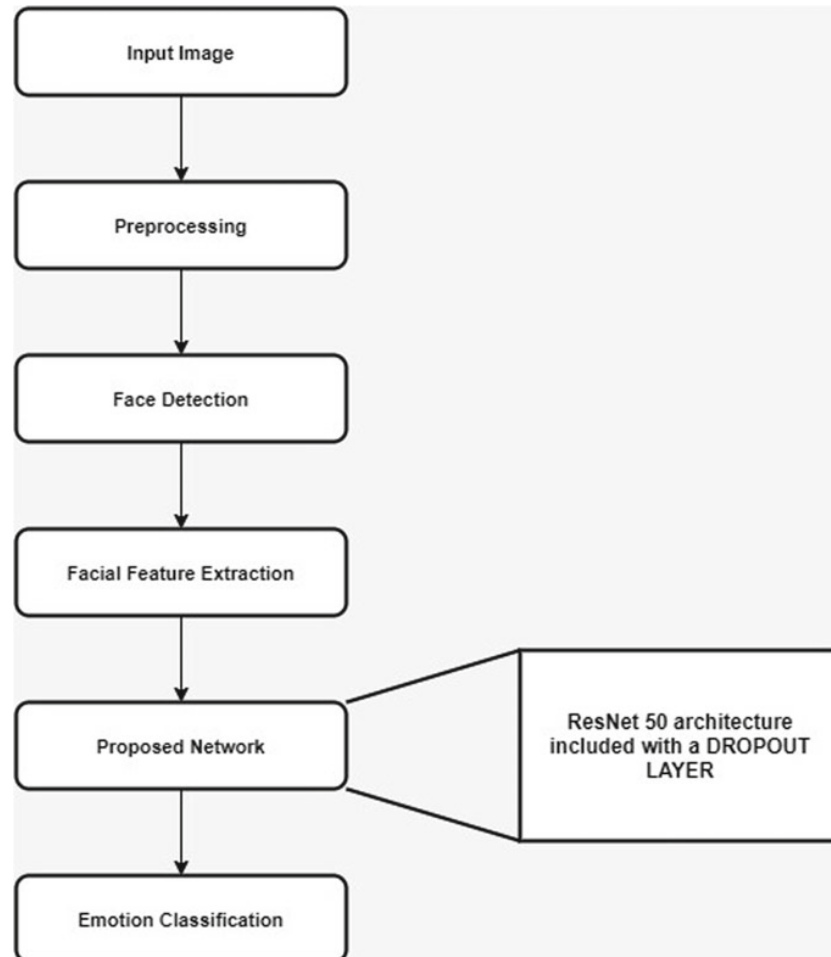


Fig. 3 Proposed model

Convolutional Neural networks are extremely good at abstract features from the face such as landmarks of face and distances between them.

4.1 Convolutional Neural Network

Convolutional Neural Network (CNN) has been pre-dominant among various deep learning models. This is a class of artificial neural networks that have produced astonishing results in the area of computer vision. A Convolutional Neural Network

is a Deep learning Model used for processing, the type of data that has a grid pattern in it. These networks are built to learn the features and patterns from the images automatically and adaptively. A CNN typically comprises three major parts, they are—convolutional block, pooling layers and fully connected networks.

CNN accepts the input in the form of tensors with the shape of an image. Then the image is passed through the convolutional layer for the abstraction of features from it with a stack of multiple mathematical operations. Further, these convolutional layers convolve the inputs and move the output to the next layer. Furthermore, Pooling layers are used to minimise the dimensions of the input by aggregating the outputs of neuron clusters of a layer into a single neuron in the next layer. Additionally, the fully connected layers are used to classify the images. The neurons present in one layer are connected to each and every neuron in another layer. The hierarchy of extracted features become more complex as the layers feed their output to another layer as input. It uses back-propagation and gradient descent for the optimisation of weights and biases.

4.1.1 Convolutional Layer

Convolutional layer is responsible for feature extraction from the given image, simply known as feature extractor. The various features in the image are captured by a set of filters in a convolutional layer. Moreover, filters are simply a group of neurons which recognises the features and patterns at different places of an image. Further, feature maps are produced by a convolutional block convolving the filters across the dimensions of an image. Hyper parameters such as depth, striding and padding are used to produce feature maps from a convolutional layer.

4.1.2 Pooling Layer

Pooling layers are introduced in a convolutional network, to convolve or reduce the dimensions of each feature map produced by the convolutional layer. Moreover, pooling layer reduces the number of parameters as a result computational cost is reduced when images are too large to compute. Although, dimensions are reduced on the feature map by reducing the parameters, the important features and information is maintained by the pooling layer. Pooling can be done in three types: average pooling, max pooling, sum pooling.

4.1.3 Fully Connected Layer

The main objective of CNN is to classify the input into its class with the help of features. The previous convolutional layers along with the pooling layer help in feature extraction and reduce the dimensionality to minimize the computational cost;

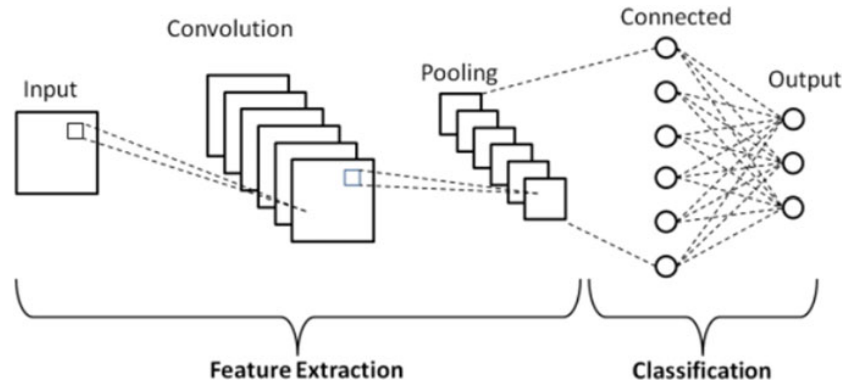


Fig. 4 Simple convolutional neural network

therefore fully connected layers are responsible for classifying the data or image into its class or category (Fig. 4).

ResNet50 architecture is employed to train the dataset ResNet, which is also known as the Residual Network. This is a convolutional neural network model which works on the concept of deep residual block. According to the universal approximation theorem, a single layer of feed forward network is used to represent any function. However, the layer might be massive and the network is prone to overfitting of the data. Therefore, our network architecture needs to go deeper. Increase in the deeper network doesn't work, resulting in a vanishing gradient problem, as the gradients are back propagated to the previous layers, rigorous multiplication operations are being conducted, gradually performance gets saturated and decreases rapidly.

A dropout is one of the regularisation techniques which prevents over fitting in the network. This dropout layer is added before the fully connected layers of ResNet50 architecture. This results in a great increase in test and train accuracy.

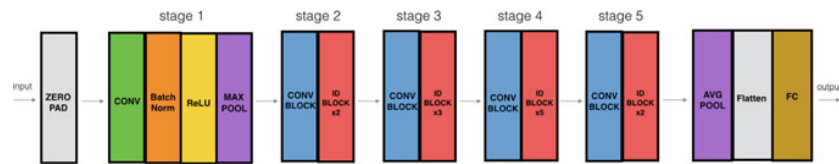
5 Experiments Analysis and Results

We have experimented with the proposed approach on the FER2013 Dataset. The obtained results are shown in Table 1 and Fig. 5. As the dataset is in grey-scale and size is 48×48 , there is no need for pre-processing the images in the dataset. The dataset is made into batches with batch size of 64. Additionally, these batches of images were trained on the ResNet model with the help of a few hyper parameters. Cross Entropy loss function is used to calculate loss from the training result and the original dataset, loss function is given below.

Table 1 Comparison of test accuracies

Method	Test accuracies (%)
AlexNet	14.9
Vgg16	38.2
ResNet50	46.6
Proposed network	55.6

Highlighted value shows the higher accuracy rate

**Fig. 5** ResNet50 architecture

$$H_p(q) = -\frac{1}{N} \sum_{i=1}^N y_i \log(p(y_i)) + (1 - y_i) \cdot \log(1 - p(y_i))$$

q is the input data and p is the obtained result.

The Optimization algorithm used is Adam optimization algorithm. Adam can be described as RMSprop and stochastic Gradient descent with momentum. Learning rate is scaled using squared gradients in the Adam optimiser. Moreover, the learning rate used is 0.01. Additionally, we have added a dropout layer before the fully connected layers to the ResNet50 which helped model to produce best results for that dataset. Furthermore, this dropout layer helped the problem of over fitting. These hyper-parameters are used in training the FER2013 dataset in an efficient manner. As the ResNet50 comprises trainable parameters, the network is completely trained on the FER2013 dataset. Lastly, we have obtained a training accuracy rate of 75.45% and test accuracy rate of 54.56%.

Figure 6 plotted between epochs: x-axis, loss: y-axis of training data. From the above graph we can observe that loss is minimum at 60th epoch. Though there are more epochs, the accuracies are oscillating among the same loss with less difference, which result in over fitting.

Table 1 depicted the test accuracies produced over different networks and proposed network. The selected networks are AlexNet, Vgg16 and ResNet50. The accuracy obtained after experimenting on these networks are 14.9%, 38.2% and 46.6% respectively. The test accuracies are gradually increasing on these networks and the proposed network has produced the highest accuracy is 55.6%.

Figure 7 shows the test results by passing custom images to the proposed network. The proposed network produced correct results over custom test images.

Fig. 6 Graph plot between loss and epochs

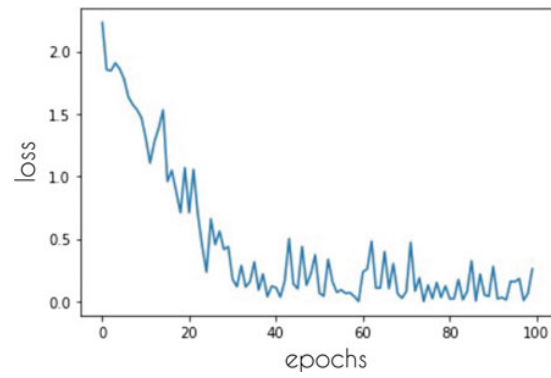


Fig. 7 Test results from proposed network

6 Conclusion

We have proposed a network by adding additional dropout layers to the actual ResNet50 architecture at required time steps. The proposed model produced better

results than the other existing networks. We have achieved a training accuracy of 75% and test accuracy of 55% using the proposed network.

The future work may include collecting and adding some more images into this FER2013 dataset so that all the emotions will be balanced. It will ensure the same number of images in each emotion class with more variety of poses.

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